Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	06 July 2026
Team ID	Individual
Project Name	India's Agricultural Crop Production Analysis (1997-2021)
Maximum Marks	4 Marks

Functional Requirements:
Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	**User Registration**	● Registration through Form (Name, Organization, Contact details) ● Registration through Gmail ● Registration as Farmer / Policy Maker / Researcher / Admin role
FR-2	**User Authentication & Confirmation**	● Confirmation via Email verification link ● Login via Email/Username & Password ● Confirmation via OTP for password reset
FR-3	**Data Upload & Management**	● Upload crop production dataset (CSV/Excel) ● View, edit, or delete uploaded records ● Validate data format before processing
FR-4	**Data Visualization & Analysis (Tableau)**	● State wise Crop Production analysis ● Year wise Crop Production trend analysis (1997-2021) ● Crop wise Production and Yield comparison ● Season wise (Kharif/Rabi) Production analysis ● District wise Production analysis ● Area vs Production vs Yield analysis ● Top producing States and Crops analysis ● Crop Production growth/decline trend over the years ● State wise Yield efficiency analysis ● Crop diversification analysis across regions
FR-5	**Report Generation**	● Generate summary reports of the analysis ● Export dashboards/reports as PDF or Image
FR-6	**Admin Dashboard**	● View all uploaded datasets and reports ● Manage user access and permissions

Non-functional Requirements:
Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	Dashboards should be simple and intuitive so that non-technical users such as farmers can easily interpret the visualized crop data.
NFR-2	**Security**	Dataset access should be restricted through authentication and role-based permissions to protect uploaded agricultural data.
NFR-3	**Reliability**	Dashboards should consistently reflect accurate and up-to-date data without errors or data loss across the 25-year dataset.
NFR-4	**Performance**	Visualizations and dashboards should load quickly even when working with large, multi-year crop production datasets.
NFR-5	**Availability**	The dashboard and reports should be accessible to users whenever needed, with minimal downtime.
NFR-6	**Scalability**	The system should be able to handle growing dataset sizes (additional years, states, or crops) and an increasing number of users without performance degradation.